

# Region-Temporal Attention Multi-Region Color Pulse Network for Contactless Stress Recognition

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**Abstract.** Stress often appears first as a physiological change before it becomes visible in facial expression or overt behavior. Facial video therefore offers an attractive route for camera-based contactless stress recognition, but the useful skin-color cues are weak, spatially uneven, and easily masked by motion, illumination, and subject appearance. Existing video methods usually follow two routes: modeling full facial frames directly, or first recovering remote physiological signals and then extracting classification features. The former may learn identity or scene shortcuts on small stress datasets, while the latter can compress away regional differences too early. This paper proposes Region-Temporal Attention Multi-Region Color Pulse Network (RT-MRCPNet). The method converts each facial video clip into multiple regional RGB temporal sequences and treats different facial areas as weak physiological observations. A shared temporal encoder extracts regional color dynamics, temporal attention selects more reliable moments within each region, and region attention adaptively fuses facial regions into a clip-level stress representation. Under a subject-independent protocol on social-stress facial video data, RT-MRCPNet achieves 88.42% ACC, 88.15% F1-score, and 91.54% AUC. It improves over the strongest lightweight video baseline by 1.89, 2.27, and 2.33 percentage points, and over the fixed multi-region fusion baseline by 4.80, 5.11, and 5.10 percentage points in ACC, F1-score, and AUC, respectively.

**Keywords:** Contactless stress recognition · Facial video · Remote photoplethysmography · Color pulse sequence · Region-temporal attention

## 1 Introduction

Stress recognition is most useful outside controlled laboratories, where remote interviews, online learning, telemedicine, and workload monitoring require sensing that is easy to deploy and minimally intrusive. Ordinary cameras provide a natural contactless signal source: facial videos contain expression, gaze, and motion, but also weak skin-color fluctuations caused by blood volume changes. Remote photoplethysmography (rPPG) studies have used such fluctuations for heart-rate estimation, HRV analysis, emotion monitoring, and stress-related inference [1–3]. Datasets such as UBFC-Phys, StressID, and ForDigitStress further

place facial videos in social-stress or digital-interaction protocols [4–6]. The resulting task is practically meaningful but difficult: stress-related color changes are weak, local, and easily masked by identity appearance, motion, and illumination.

Current facial-video methods can lose useful evidence between full-frame visual modeling and global physiological summarization. Full-frame models preserve rich appearance detail, and networks such as DeepPhys, PhysNet, MTTS-CAN, and EfficientPhys show that deep models can learn remote physiological representations [7–10]. However, small stress datasets make such models vulnerable to shortcuts related to identity, background, skin tone, or task setting. Traditional pipelines instead recover a whole-face rPPG signal and classify HR/HRV features [1, 3], but this can collapse differences among the forehead, cheeks, nose, and chin too early. Since rPPG quality varies across facial regions [11–13], contactless stress recognition needs a compact representation that retains regional evidence.

Aggregation is another challenge because frame and region reliability changes within each stress video. Speech, arithmetic, blinking, head motion, occlusion, reflection, and asymmetric lighting may corrupt different regions at different moments. Fixed averaging therefore mixes discriminative responses with noisy segments. Attention mechanisms can assign input-dependent weights to time positions and regions, and prior physiological-video methods have used attention for remote physiological measurement or stress recognition [7, 9, 14]. For small stress datasets, however, a lightweight attention design is preferable to large Transformer-style models.

We propose Region-Temporal Attention Multi-Region Color Pulse Network (RT-MRCPNet). The method treats facial regions as multiple weak physiological observations, learns stress-related representations from regional RGB dynamics, uses temporal attention to select reliable moments within each region, and uses region attention to fuse discriminative facial areas. The contributions are: (1) a video-only stress formulation based on multi-region color-pulse sequences; (2) a compact region-temporal attention network that separates temporal reliability from regional contribution; and (3) subject-independent validation on UBFC-Phys, where RT-MRCPNet achieves 88.42% ACC, 88.15% F1-score, and 91.54% AUC, outperforming full-face RGB, single-region, fixed-fusion, hand-crafted rPPG, and lightweight video baselines.

## 2 Related Work

### 2.1 Stress Recognition and Multimodal Data

Stress recognition has long relied on contact physiology such as ECG, EDA, PPG, respiration, and skin temperature because these signals are close to stress-response mechanisms [15–17]. HRV reflects autonomic regulation, EDA indicates sympathetic arousal, and respiration changes with cognitive load, but sensors require attachment, synchronization, and user cooperation [16, 18]. Recent

datasets therefore move toward more realistic multimodal settings: UBFC-Phys records facial videos with wrist BVP and EDA during rest, speech, and arithmetic [4]; StressID adds audio, ECG, EDA, and respiration under emotional, cognitive, and speaking stimuli [5]; and ForDigitStress studies digital interviews with video, gaze, audio, PPG, and EDA [6]. These datasets motivate contactless stress recognition but also show that high-quality stress data remain small and protocol-dependent.

## 2.2 Contactless Stress Recognition and Facial Color Cues

Early contactless studies used facial video to estimate remote physiological indicators before classification. McDuff et al. [1] estimated heart rate, respiration, and HRV for cognitive stress classification; Nikolaiev et al. [3] recovered HRV indicators from webcam rPPG; and Benezeth et al. [2] explored remote HRV for emotion monitoring. More recent work moves toward direct video learning: Xu et al. [14] used multi-task learning with peak attention for facial video stress recognition, while SympCam [19] and related work [20] estimate sympathetic arousal from face and hand videos. These studies show that stress recognition is related to rPPG but not equivalent to recovering one clean pulse waveform; discriminative cues may appear as local color dynamics, short temporal peaks, or task-induced speech and motion.

Traditional rPPG methods such as CHROM [21], POS [22], and LGI [23] show that RGB trajectories contain weak cardiovascular information. DeepPhys [7], PhysNet [8], MTTS-CAN [9], and EfficientPhys [10] further learn physiological evidence from video variation. Their targets are usually vital-sign estimation, whereas stress recognition requires representations that turn remote color cues into discriminative task-related features.

## 2.3 Regional Modeling and Attention

Facial region selection is central to rPPG quality because local blood perfusion, skin texture, pose, lighting, and motion vary across ROIs. Multi-scale and multi-ROI methods improve robust heart-rate estimation [11], recent studies emphasize that region selection remains open [13], and GraphPhys models relationships among facial regions with a graph neural network [12]. Attention is also widely used to suppress background or unreliable frames and to emphasize informative waveform segments. However, existing attention designs mainly serve full-video representation learning or remote physiological reconstruction. This work instead asks whether compact regional color sequences, optimized directly for stress classification, can improve subject-independent recognition through two lightweight attention stages: temporal aggregation within each ROI and contribution fusion across ROIs.

### 3 Method

#### 3.1 Problem Definition

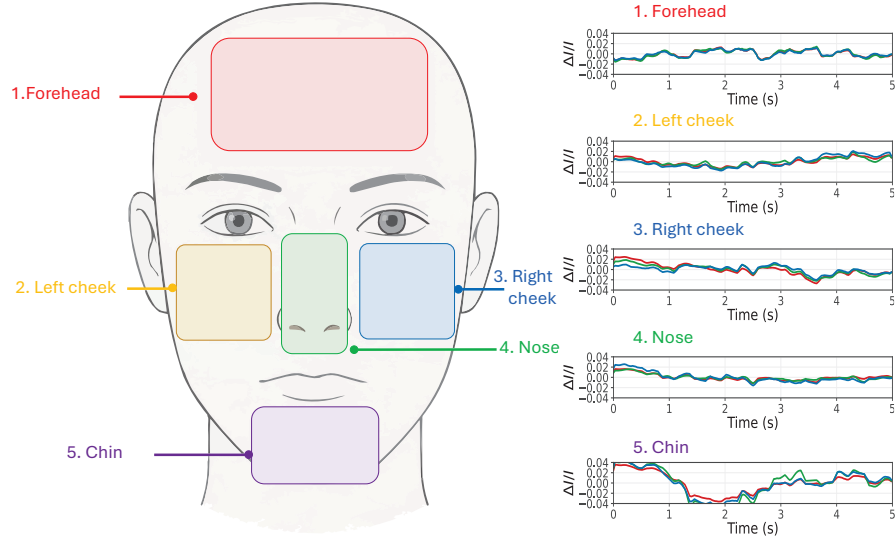
Given a facial video clip  $V = \{I_1, I_2, \dots, I_T\}$ , the goal is to predict  $y \in \{0, 1\}$ , where 0 denotes non-stress and 1 denotes stress. The video is represented as

$$X \in \mathbb{R}^{K \times T \times C}, \quad (1)$$

where  $K$  is the number of ROIs,  $T$  is the clip length, and  $C = 3$  is the number of RGB channels. For a batch,  $X \in \mathbb{R}^{B \times K \times T \times C}$ .

#### 3.2 Multi-Region Color Pulse Representation

For each frame, the face is detected, cropped, and resized so that fixed ROIs remain geometrically consistent. Figure 1 shows the five regions used in this paper: forehead, left cheek, right cheek, nose, and chin. They cover common rPPG skin areas while retaining different reliability patterns: cheeks often provide clear color dynamics, the forehead is stable but can be affected by hair, the nose is sensitive to highlights, and the chin is affected by speech.



**Fig. 1.** Five facial regions used by the multi-region color pulse representation.

For the  $k$ -th region at time  $t$ , the regional RGB vector is obtained by average pooling over all pixels in the region:

$$x_{k,t} = \frac{1}{|\Omega_k|} \sum_{p \in \Omega_k} I_t(p), \quad (2)$$

where  $\Omega_k$  denotes the pixel set of the  $k$ -th ROI. The sequence of the  $k$ -th ROI is

$$X_k = [x_{k,1}, x_{k,2}, \dots, x_{k,T}]. \quad (3)$$

All regional sequences are stacked to obtain  $X \in \mathbb{R}^{K \times T \times C}$ , where  $C = 3$  corresponds to the RGB channels.

Temporal standardization is applied independently for each ROI and channel:

$$\hat{X}_{k,:,c} = \frac{X_{k,:,c} - \mu_{k,c}}{\sigma_{k,c} + \epsilon}, \quad (4)$$

where  $\hat{X}$  is the standardized sequence,  $c$  indexes the color channel,  $\mu_{k,c}$  and  $\sigma_{k,c}$  are the clip-wise mean and standard deviation, and  $\epsilon$  prevents division by zero. This removes most spatial redundancy while preserving regional color dynamics.

### 3.3 Region-Wise Temporal Encoder

Each ROI sequence is encoded by a shared one-dimensional temporal encoder. Given  $\hat{X} \in \mathbb{R}^{B \times K \times T \times C}$ , we reshape it into  $\tilde{X} \in \mathbb{R}^{BK \times C \times T}$  and apply stacked one-dimensional convolutional blocks:

$$H = E_\theta(\tilde{X}), \quad (5)$$

After dimension permutation,  $H \in \mathbb{R}^{BK \times T \times D}$ , where  $D$  is the hidden dimension. Let  $h_{k,t} \in \mathbb{R}^D$  denote the feature for region  $k$  at time  $t$ . Sharing the encoder across ROIs captures common low-level color dynamics while reducing parameters and overfitting. Short temporal kernels model local changes without repeatedly processing full spatial image grids.

### 3.4 Temporal Attention

For each ROI sequence, temporal attention predicts a scalar score for each encoded time position:

$$e_{k,t} = \mathbf{w}_t^\top \tanh(\mathbf{W}_t h_{k,t}), \quad (6)$$

where  $\mathbf{W}_t$  is the feature projection matrix in temporal attention and  $\mathbf{w}_t$  is the weight vector used to produce a scalar score. The normalized temporal weight is

$$\alpha_{k,t} = \frac{\exp(e_{k,t})}{\sum_{\tau=1}^T \exp(e_{k,\tau})}. \quad (7)$$

The ROI-level representation is computed as

$$r_k = \sum_{t=1}^T \alpha_{k,t} h_{k,t}. \quad (8)$$

Each region therefore has its own temporal weighting pattern, allowing speech, motion, or illumination artifacts to be suppressed before region fusion.

### 3.5 Region Attention

After temporal aggregation, the ROI features are denoted as  $r_1, \dots, r_K$ . Region attention computes a contribution score for each ROI:

$$s_k = \mathbf{w}_r^\top \tanh(\mathbf{W}_r r_k), \quad (9)$$

where  $\mathbf{W}_r$  is the feature projection matrix in region attention and  $\mathbf{w}_r$  is the weight vector used to produce a scalar region score. The region weights are then obtained by softmax:

$$\beta_k = \frac{\exp(s_k)}{\sum_{j=1}^K \exp(s_j)}. \quad (10)$$

The attention-based clip representation is

$$z_{\text{attn}} = \sum_{k=1}^K \beta_k r_k. \quad (11)$$

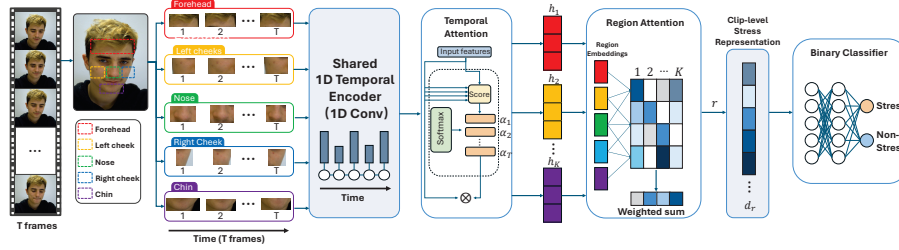
For stable training, an average regional representation is added as a residual term:

$$z = z_{\text{attn}} + \frac{1}{K} \sum_{k=1}^K r_k. \quad (12)$$

Finally,  $z$  is passed to a two-layer MLP classifier to obtain logits:

$$o = \text{MLP}(z). \quad (13)$$

Figure 2 summarizes the RT-MRCPNet pipeline.



**Fig. 2.** Architecture of RT-MRCPNet.

### 3.6 Training Objective

The model is trained with cross-entropy loss. Let  $p_c = \text{softmax}(o)_c$  denote the predicted probability of class  $c$ :

$$\mathcal{L}_{cls} = - \sum_{c=1}^2 y_c \log p_c, \quad (14)$$

where  $y_c$  is the one-hot label at class  $c$ . We use only classification supervision, keeping the comparison focused on regional RGB dynamics and attention-based aggregation.

## 4 Experiments

### 4.1 Dataset and Label Definition

The primary experiment uses UBFC-Phys-style social stress facial videos. T1 is rest and is labeled non-stress ( $y = 0$ ), while T2 and T3 are speech and arithmetic tasks and are labeled stress ( $y = 1$ ).

### 4.2 Subject-Independent Protocol

All experiments use subject-independent splits: clips from the same participant do not appear in both training and testing sets. We use 70% of subjects for training, 15% for validation, and 15% for testing; validation F1-score selects the checkpoint.

### 4.3 Preprocessing and Implementation Details

Faces are cropped to  $128 \times 128$ , divided into clips of  $T = 160$  frames, and converted into five standardized ROI RGB sequences. The hidden dimension is 128. We train for 100 epochs with AdamW, learning rate  $10^{-3}$ , and batch size 64 on one NVIDIA RTX 3090 using PyTorch 2.1.

### 4.4 Evaluation Metrics

We report accuracy (ACC), F1-score, and AUC, treating stress as the positive class.

### 4.5 Comparison Experiments

All methods use the same input source, subject-independent split, clip construction, and evaluation code whenever applicable. Full-face RGB, single-region RGB, and fixed multi-region fusion are baselines constructed in this paper; traditional rPPG methods, video temporal models, and rPPG backbones follow existing methods or structures.

- Full-face RGB temporal network. The five regional sequences are averaged into a single face-level RGB sequence before temporal encoding.
- Single-region RGB temporal networks. Forehead, left cheek, right cheek, nose, and chin are evaluated separately to reveal regional performance differences.
- Multi-region fixed fusion. The model keeps multiple regional RGB sequences but uses non-adaptive concatenation or average fusion instead of attention.

- Traditional rPPG features with classical classifiers. POS or CHROM-style color traces [22, 21] are converted into HR/HRV features, followed by support vector machine or random forest classifiers.
- Lightweight video temporal baselines. A compact CNN-LSTM or 3D CNN receives cropped face clips and predicts the stress label directly. The CNN-LSTM-style baseline follows the idea of CCT-LSTM for remote stress estimation [24], while the 3D CNN baseline follows classic three-dimensional convolutional video modeling [25].
- Existing rPPG backbones with a stress head. Representative rPPG models, such as PhysNet [8] or EfficientPhys [10], are adapted by replacing the signal-regression output with a stress classification head.
- RT-MRCPNet. The proposed model uses multi-region RGB sequences with temporal attention and region attention.

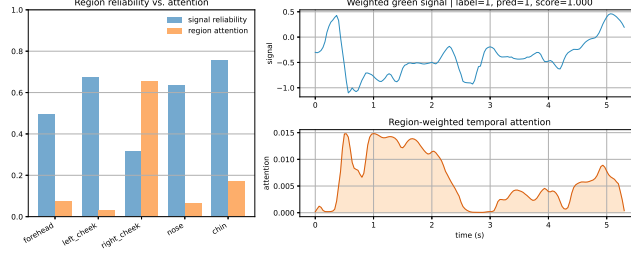
The resulting performance is summarized in Table 1.

**Table 1.** Controlled comparison under the UBFC-Phys subject-independent protocol. All metrics are reported in percentage (%).

| Method                                 | Input                                    | ACC          | F1           | AUC          |
|----------------------------------------|------------------------------------------|--------------|--------------|--------------|
| Full-face RGB temporal network         | Averaged face-level RGB sequence         | 81.42        | 80.85        | 84.51        |
| Forehead-only temporal network         | Forehead RGB sequence                    | 79.80        | 78.64        | 82.05        |
| Left-cheek-only temporal network       | Left cheek RGB sequence                  | 82.55        | 81.72        | 85.23        |
| Right-cheek-only temporal network      | Right cheek RGB sequence                 | 82.14        | 81.48        | 84.81        |
| Nose-only temporal network             | Nose RGB sequence                        | 76.40        | 75.12        | 79.55        |
| Chin-only temporal network             | Chin RGB sequence                        | 77.85        | 76.50        | 80.12        |
| Multi-region fixed fusion              | Five ROI RGB sequences, fixed fusion     | 83.62        | 83.04        | 86.44        |
| rPPG features + support vector machine | Handcrafted HR/HRV features              | 78.50        | 77.25        | 81.33        |
| rPPG features + random forest          | Handcrafted HR/HRV features              | 80.25        | 79.52        | 83.10        |
| Lightweight CNN-LSTM                   | Cropped face clips                       | 84.20        | 83.55        | 87.12        |
| Lightweight 3D CNN                     | Cropped face clips                       | 85.15        | 84.78        | 88.02        |
| PhysNet/EfficientPhys stress head      | + Face clips or rPPG-oriented features   | 86.53        | 85.88        | 89.21        |
| RT-MRCPNet                             | Five ROI RGB sequences, attention fusion | <b>88.42</b> | <b>88.15</b> | <b>91.54</b> |

Table 1 and Fig. 3 show that RT-MRCPNet performs best among the RGB-sequence, rPPG-feature, and lightweight video baselines. The cheek-only models outperform the nose and chin, confirming uneven regional signal quality. Full-face





**Fig. 3.** Attention reliability visualization.

averaging mixes stable areas with regions affected by speech, reflection, or occlusion, while fixed multi-region fusion still assumes equal reliability. RT-MRCPNet improves by preserving regions as separate observations and adaptively selecting reliable moments and facial areas. This avoids premature compression into one global rPPG signal and avoids training a heavy full-frame video model, making the method better suited to small subject-independent stress datasets.

#### 4.6 Ablation Study

The ablation study uses the same protocol as Table 1 and removes individual attention modules from RT-MRCPNet.

**Table 2.** Ablation study of RT-MRCPNet. All metrics are reported in percentage (%).

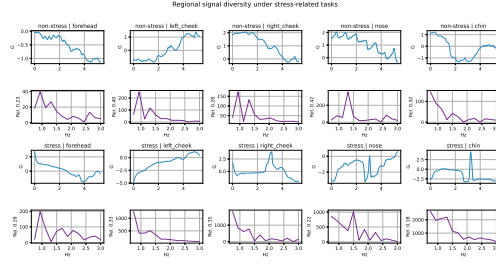
| Variant                 | ACC          | F1           | AUC          |
|-------------------------|--------------|--------------|--------------|
| Without attention       | 83.62        | 83.04        | 86.44        |
| Temporal attention only | 86.15        | 85.52        | 88.90        |
| Region attention only   | 85.34        | 84.71        | 88.22        |
| Full RT-MRCPNet         | <b>88.42</b> | <b>88.15</b> | <b>91.54</b> |

The two attention stages are complementary. Without attention, the model uses fixed fusion and cannot handle short disturbances or regional quality differences. Temporal attention down-weights unstable moments within each ROI, while region attention raises the contribution of reliable facial areas. The full model performs best, indicating that stress-related evidence is selective in both time and region. Figure 4 further illustrates the diversity of regional color sequences.

#### 4.7 Complexity Analysis

For a video of length  $T$  and resolution  $H \times W$ , a full-frame model receives

$$\mathcal{O}(THWC), \quad (15)$$

**Fig. 4.** Regional signal diversity visualization.

After preprocessing, RT-MRCPNet keeps only  $K$  regional mean-color sequences:

$$\mathcal{O}(KTC). \quad (16)$$

Since  $K \ll HW$ , this removes most spatial redundancy before learning.

Let  $D$  be the hidden dimension,  $q$  the kernel size, and  $L$  the number of encoder layers. The dominant encoding complexity is

$$\mathcal{O}(BKLTqD^2), \quad (17)$$

where  $B$  is the batch size. Temporal and region attention add

$$\mathcal{O}(BKTD + BKD), \quad (18)$$

which is small compared with full-frame video modeling. Computation grows mainly with  $K$  and  $T$ , rather than image area  $HW$ . Table 3 reports empirical complexity.

**Table 3.** Model complexity comparison.

| Method                                     | Parameters | FLOPs  | Inference Time |
|--------------------------------------------|------------|--------|----------------|
| Full-frame convolutional recurrent network | 12.54M     | 15.20G | 35.42 ms       |
| Full-frame 3D convolutional neural network | 33.25M     | 38.64G | 42.15 ms       |
| Full-face RGB temporal network             | 0.82M      | 0.05G  | 1.25 ms        |
| RT-MRCPNet                                 | 0.94M      | 0.25G  | 2.38 ms        |

## 5 Conclusion

This paper presents RT-MRCPNet, a lightweight region-temporal attention network for contactless stress recognition from facial videos. It converts facial clips into multi-region RGB temporal sequences, models local color dynamics with

a shared encoder, and aggregates information through temporal and region attention. Under a subject-independent protocol, RT-MRCPNet achieves 88.42% ACC, 88.15% F1-score, and 91.54% AUC. Ablation and complexity analyses show complementary gains from the two attention stages with low computational cost. Future work will explore adaptive ROI definitions, finer-grained stress labels, and multimodal fusion.

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